ETH zürich

Master of Advanced Studies
Applied Information & Data Science

Local Interpretability LIME

From Data to Solutions
ETH Zürich · 16. July 2026

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Agenda

1. **Motivation**
   Clever Hans · don't trust performance blindly
2. **Interpretability**
   what it is, why it matters, when it doesn't
3. **A taxonomy of methods**
   by-design vs post-hoc · where LIME sits
4. **From CP & ICE to LIME**
   the limitation that forces the jump
5. **LIME in depth**
   intuition, algorithm, the objective, data types
6. **Limitations**
   kernel width, instability → bridge to SHAP
7. **Local → global: SP-LIME**
   2 trust problems, the model-choice study

01

Motivation

- A high score is not a reason.
- Models can be right for the wrong reasons.

The original "Clever Hans"

A cautionary tale from 1900s Berlin

> *The horse that could "do arithmetic" … until someone controlled the setup and found it was reading its trainer's body language.*

The claim: Hans tapped out answers to sums, dates, musical intervals.

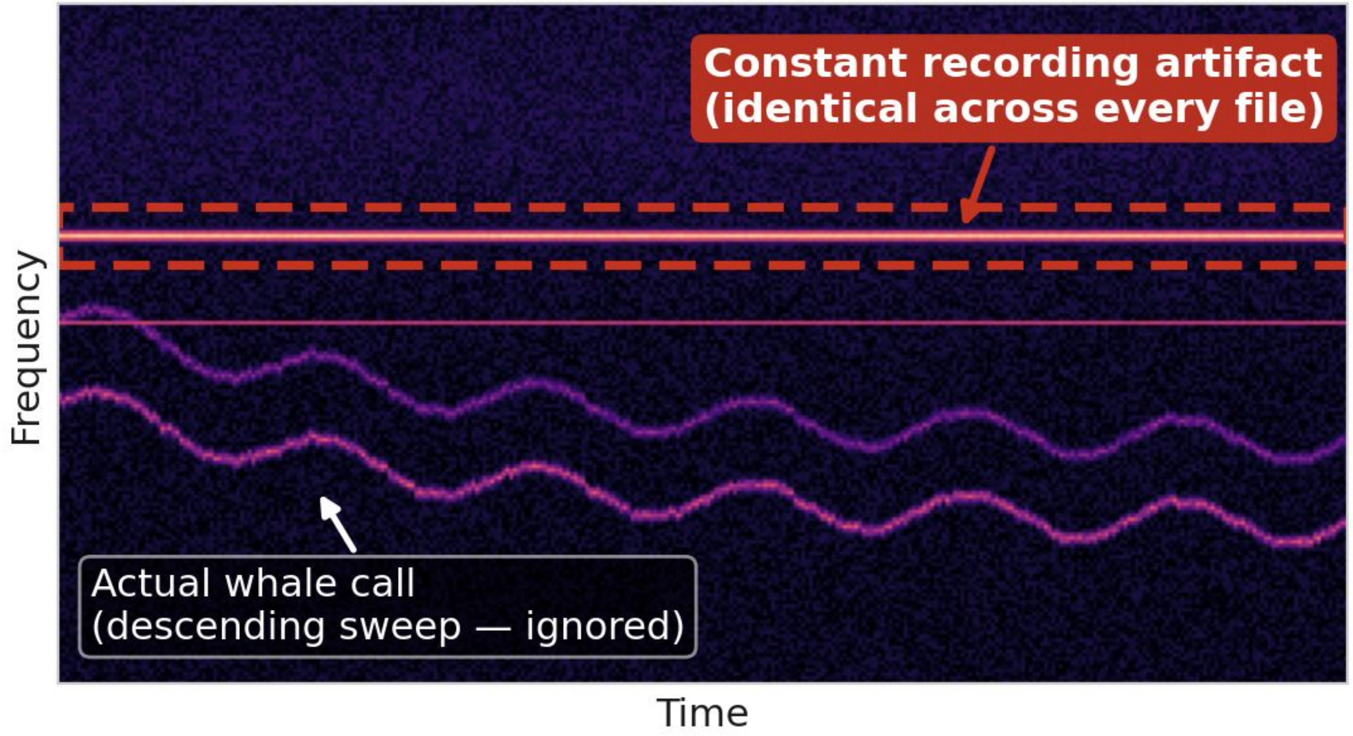
The reality: he had learned to stop tapping on subtle, unconscious cues from whoever knew the answer.

The lesson: correct outputs, wrong mechanism and no one noticed until the experiment was designed to catch it.

The whale that wasn't there

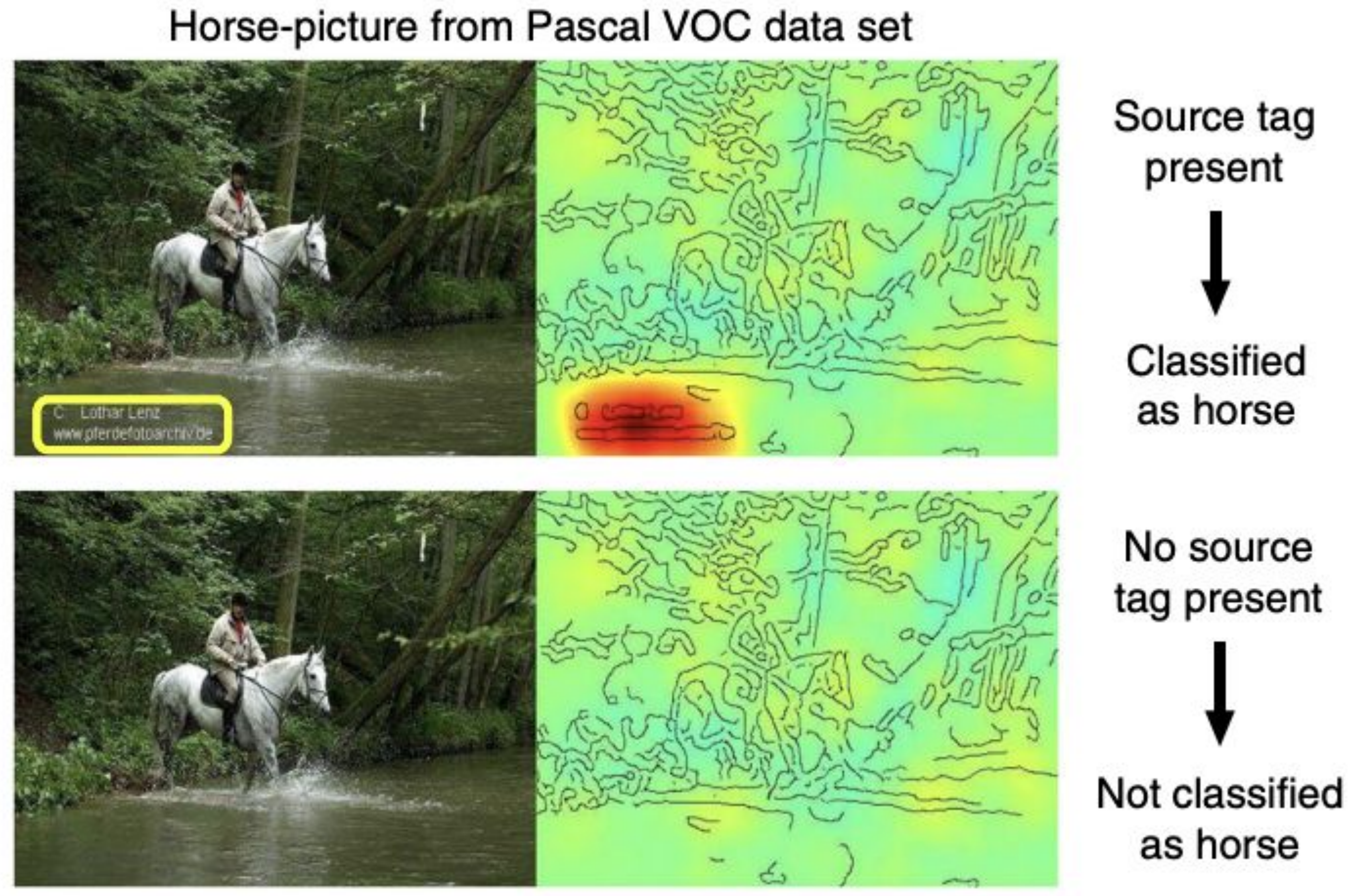
A whale-call detector scored high on the leaderboard, then failed on new recordings.

It had locked onto a constant artifact present in every training clip, ignoring the actual call.



Source: *DeLMA & Cukierski, 2013 (Kaggle Whale Detection)*

Reading the caption, not the image



A classifier appeared to recognise the object, but its evidence sat on a source watermark printed in the corner.

Remove the tag and the prediction falls apart.

Source: *Lapuschkin et al., 2019 (Nature Communications)*

A snow detector in disguise

The now-classic LIME example: a husky is called a “wolf.”

The explanation reveals the model looked at the snowy background, not the animal.

(a) Husky classified as wolf (b) Explanation

Source: *Ribeiro, Singh & Guestrin, 2016 (the LIME paper)*

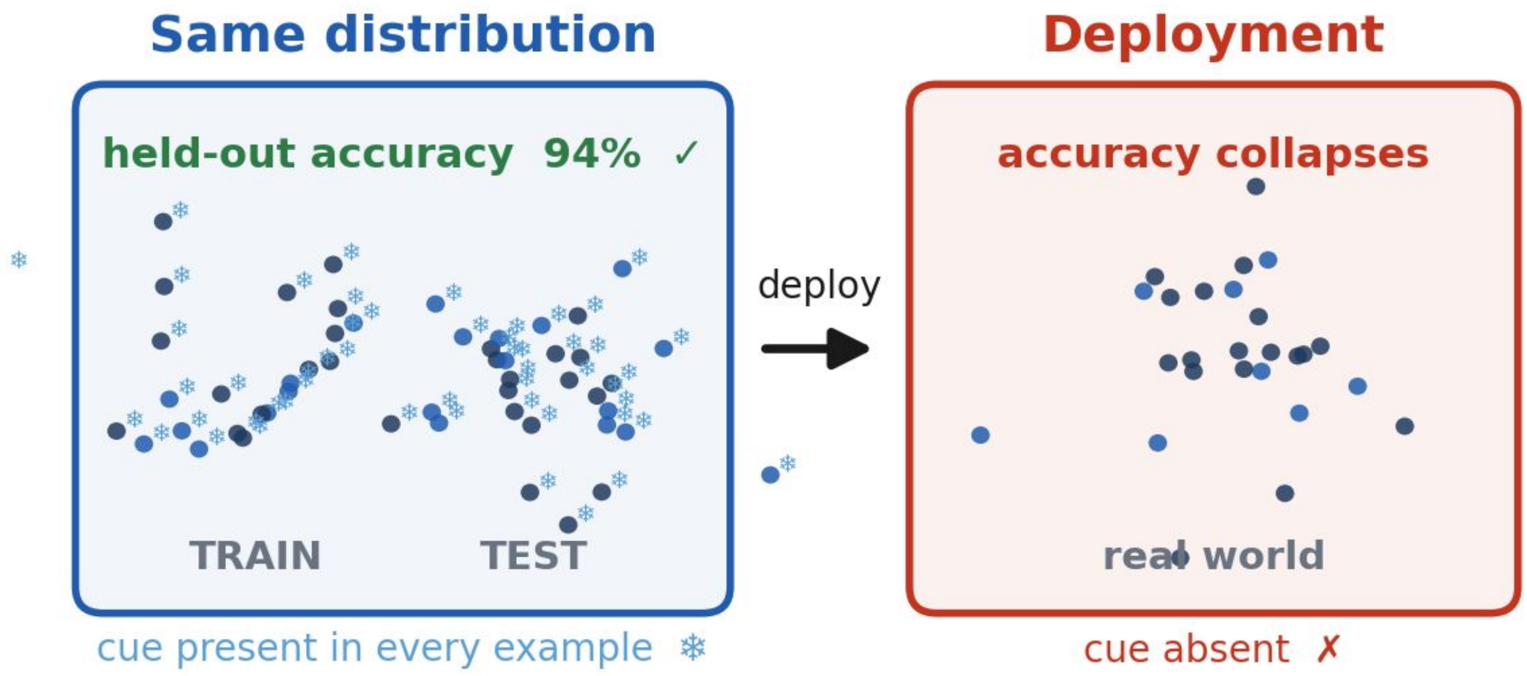
Why held-out accuracy isn’t enough

- **Same distribution**
  A held-out split only tests the distribution the model was trained on.

- **Shortcuts survive**
  Spurious cues (snow, captions, artifacts) are present at test time too.

The blind spot only opens after deployment.

In deployment the cue disappeared and so did the accuracy.



From WHAT to WHY

WHAT

0.93

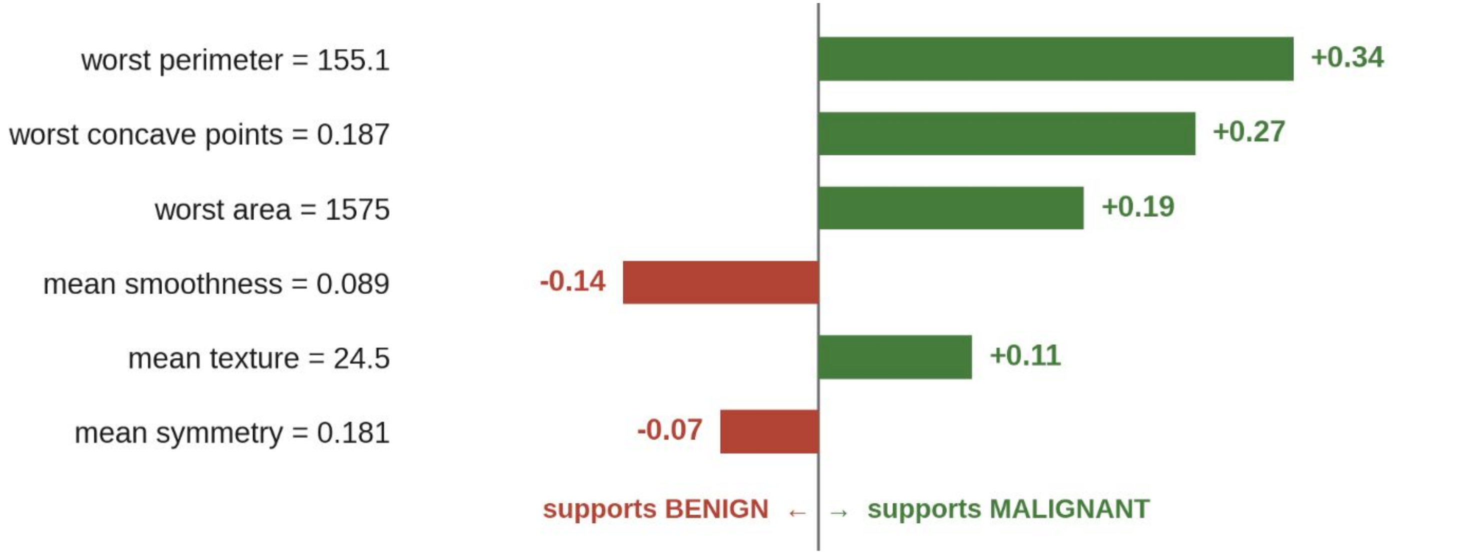
probability MALIGNANT

prediction but no reason

→

WHY which inputs drove it, and by how much

each bar: the effect of keeping that feature at the patient's value



Catch spurious correlations before deployment

Audit for bias

Build justified trust

02

What is interpretability?

- Definitions, and the fuzzy line to “explainability”.
- Why it matters — and when it doesn’t.

What is interpretability?

Two working definitions

> *The degree to which a human can understand the cause of a decision.*
>
> Miller (2017)

> *A method is interpretable if a user can correctly and efficiently predict the method's results.*
>
> Kim, Khanna & Koyejo (2016)

Interpretability vs. Explainability

I Interpretability

- A property of the model itself.
- Intrinsic transparency
- e.g. a short decision tree, a linear model.

E Explainability (XAI)

- A post-hoc artifact about a model's output.
- An explanation justifies a decision after the fact.
- e.g. LIME / SHAP on top of a black box.

In practice the terms are used almost interchangeably.

When interpretability and when not

When we need interpretability

- **Debugging** → catch spurious shortcuts
- **Trust** → justify acting on a high-stakes prediction
- **Fairness & audit** → check the model isn't using protected attributes
- **Science** → turn a predictive model into insight
- **Regulation** → GDPR-style "right to an explanation", credit, health

When we don't

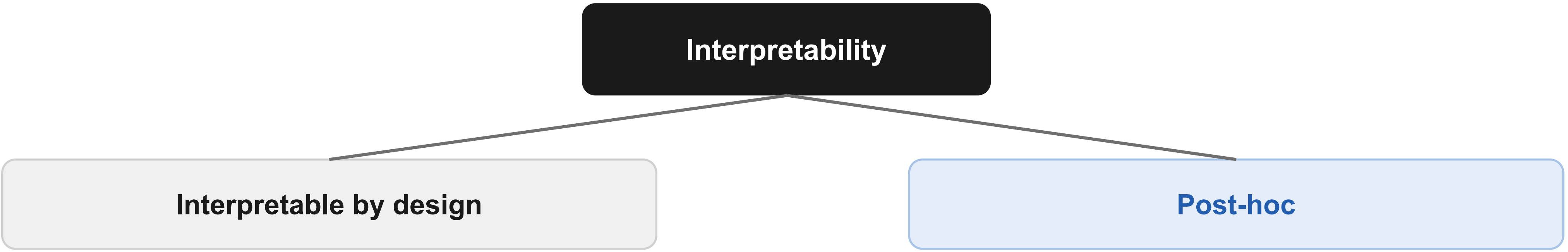
- Low-stakes decisions with no cost of error.
- Very well-studied, trusted problems.

03

A taxonomy of methods

A taxonomy of interpretability methods

LIME = post-hoc · model-agnostic · local



A taxonomy of interpretability methods

LIME = post-hoc · model-agnostic · local

- **Interpretability**
  - **Interpretable by design**
    - Linear regression
    - Logistic regression
    - Decision trees
  - **Post-hoc**

A quick recap

Linear regression

$\hat{y} = \beta_0 + \Sigma\beta_i x_i$

Prediction is a weighted sum.
Each β_i is the feature's marginal effect
read the weight, know the contribution.

Logistic regression

$\text{log-odds} = \beta_0 + \Sigma\ \beta_i\ x_i$

Same additive structure, now on the log-odds. Each weight moves the odds up or down by a fixed factor.

Decision trees

`if … then … else`

The prediction is a readable path of if-then splits from root to leaf.

Cost of transparency: these models trade some accuracy on complex data for a mechanism a human can read.

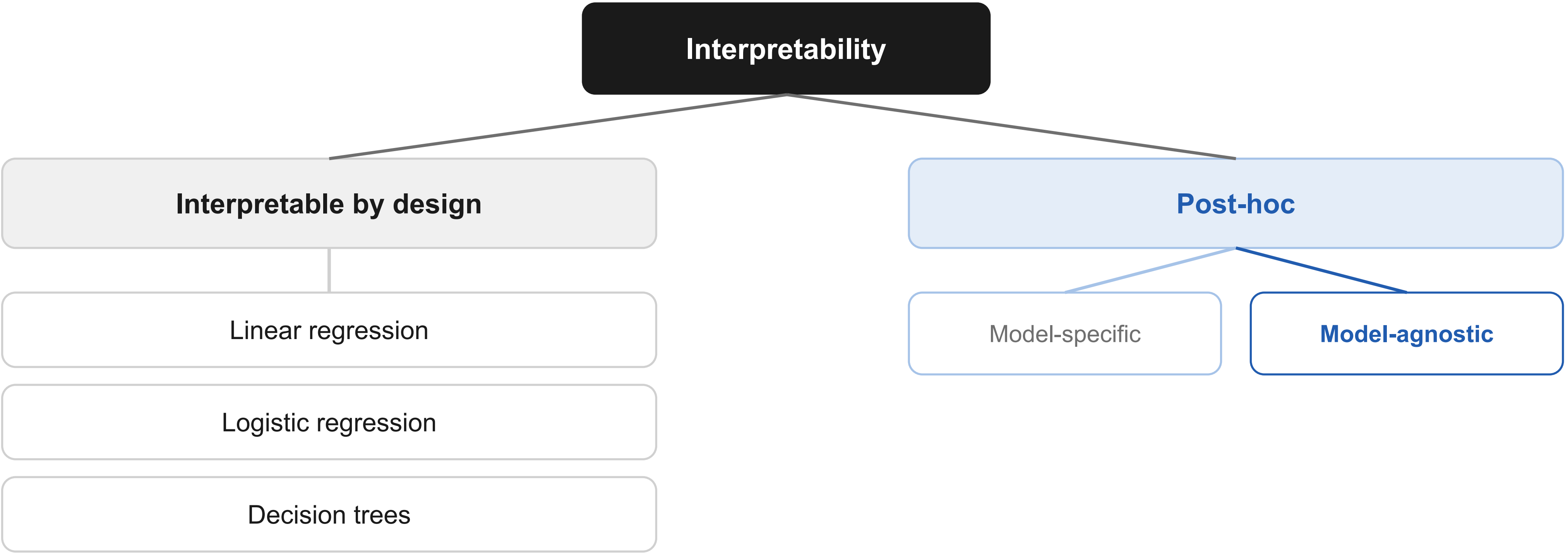
A taxonomy of interpretability methods

Interpretability

- **Interpretable by design**
  - Linear regression
  - Logistic regression
  - Decision trees
- **Post-hoc**

A taxonomy of interpretability methods

LIME = post-hoc · model-agnostic · local



The classic models on the left are interpretable in themselves. LIME instead sits on top of any black box and explains it after the fact.

A taxonomy of interpretability methods

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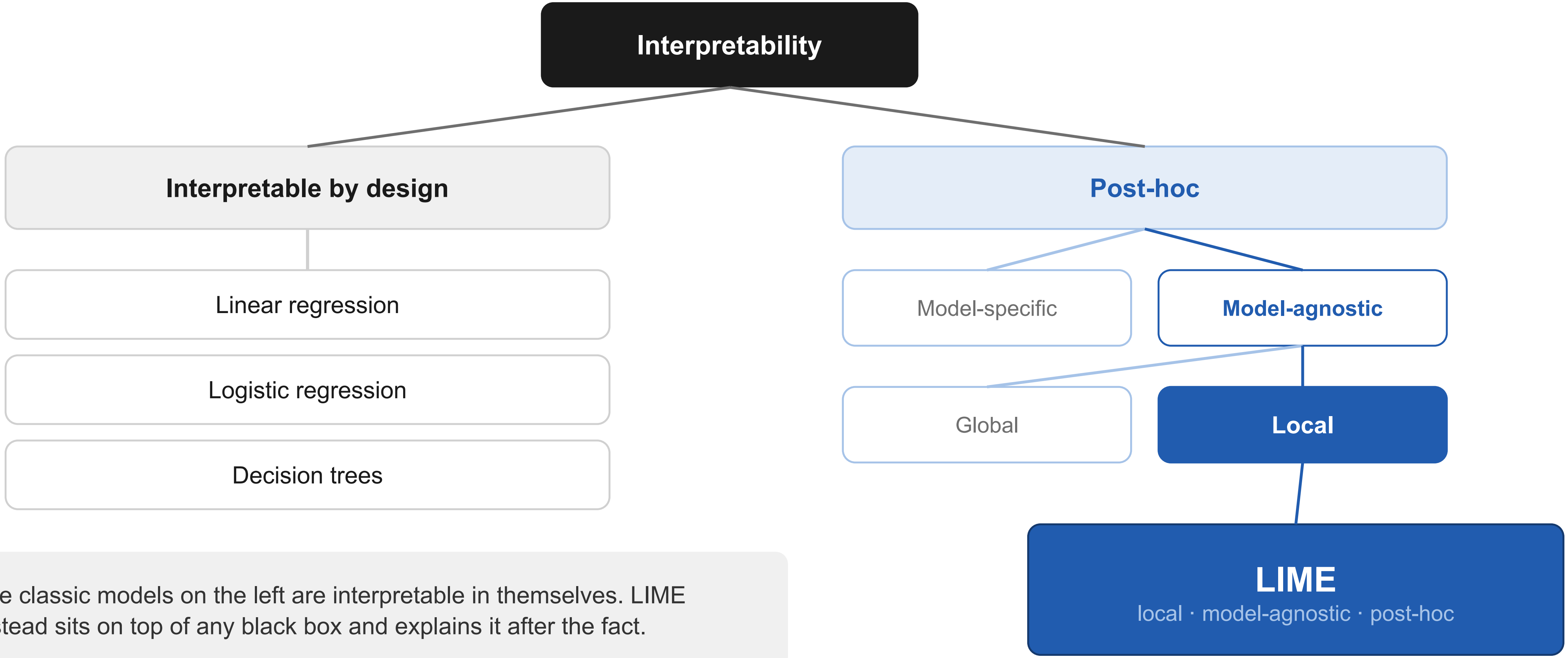

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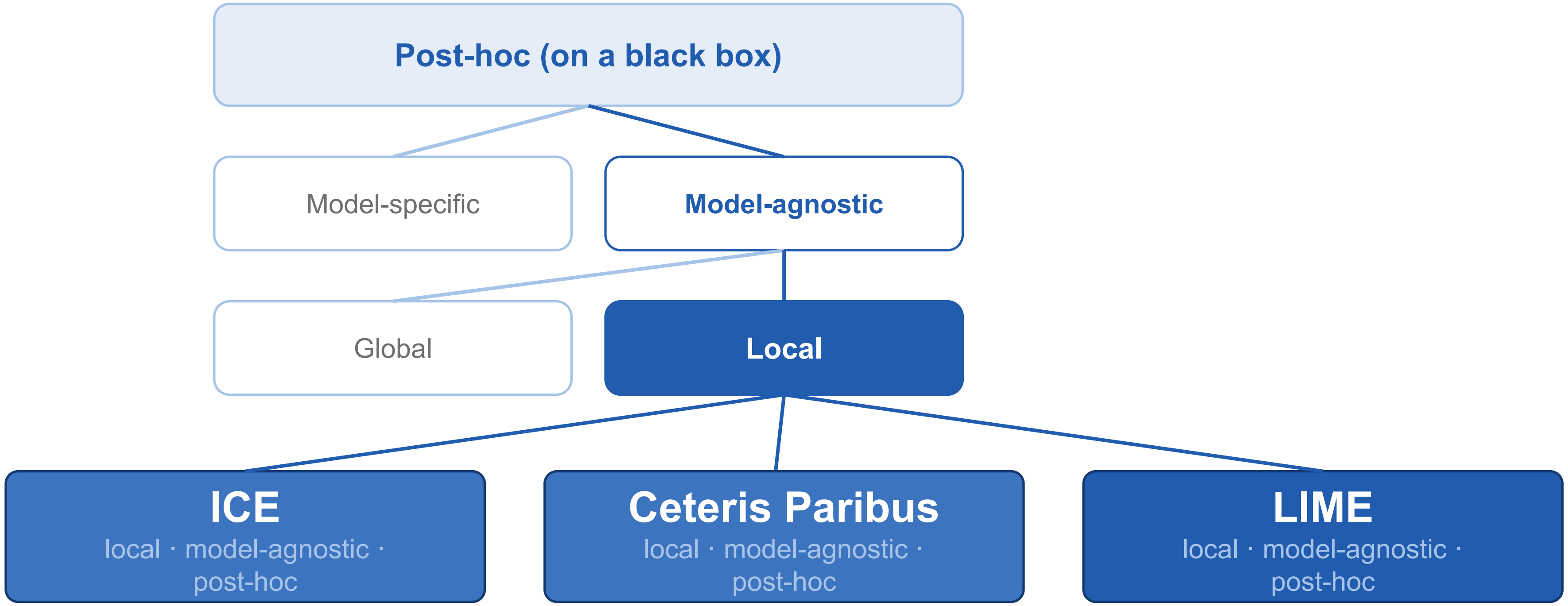


04

From ceteris paribus & ICE to LIME

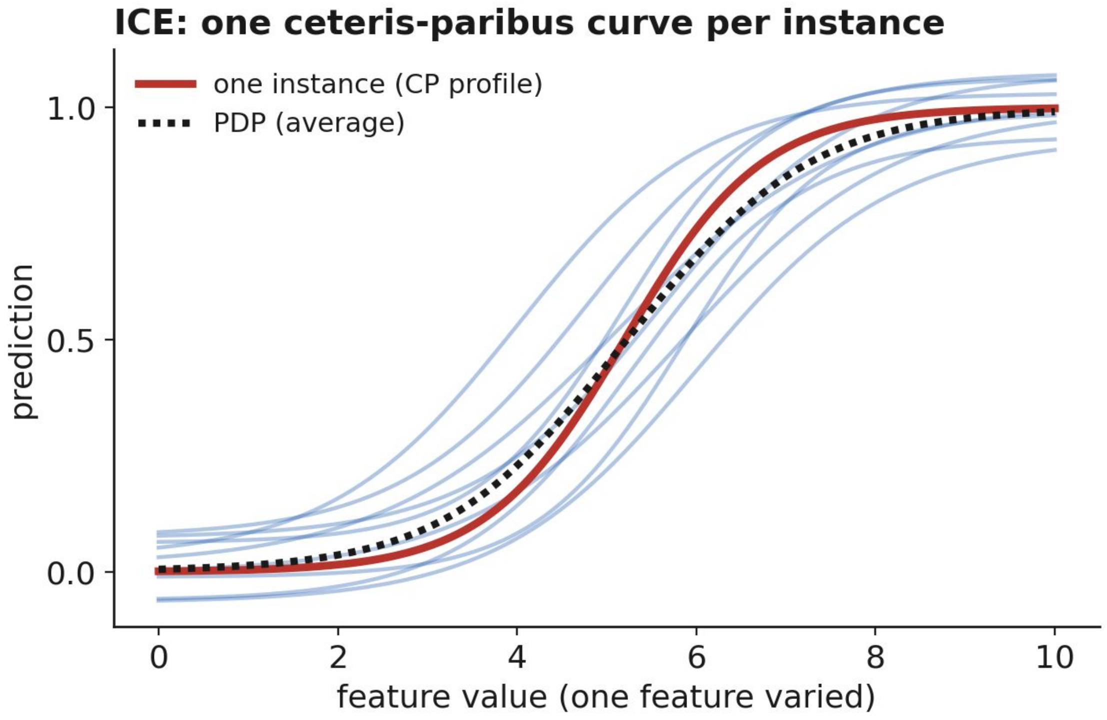
- Both vary one feature at a time.
- That limitation is exactly what LIME removes.

Ceteris-paribus & ICE plots

One feature at a time

Ceteris paribus: vary one feature, hold the rest fixed, watch the prediction move.
ICE: one such curve per instance, stacked
Great for: seeing how a single feature drives the model.

But: it isolates *one axis*. To reason about a single prediction you'd need one plot per feature …

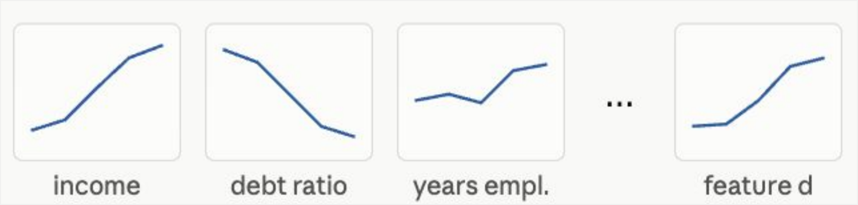


From CP / ICE to LIME

2 Limitations

1.d plots, one prediction. With many features we get no single, compact answer to “why this output?”

E.g Loan Applicant (d-features)

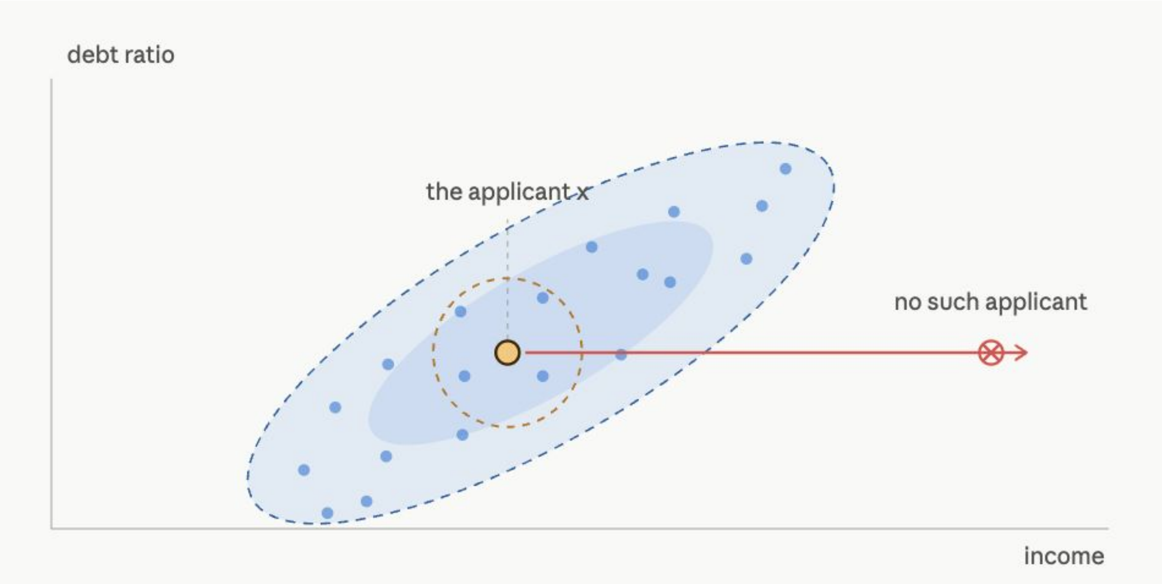


d features, d plots for 1 decision

2.Off-manifold. Holding correlated features fixed while sweeping one creates unrealistic, impossible instances the curve leaves the data.

The LIME idea

What if, instead of one axis at a time, we perturb everything at once in a small neighborhood and fit one simple model there?

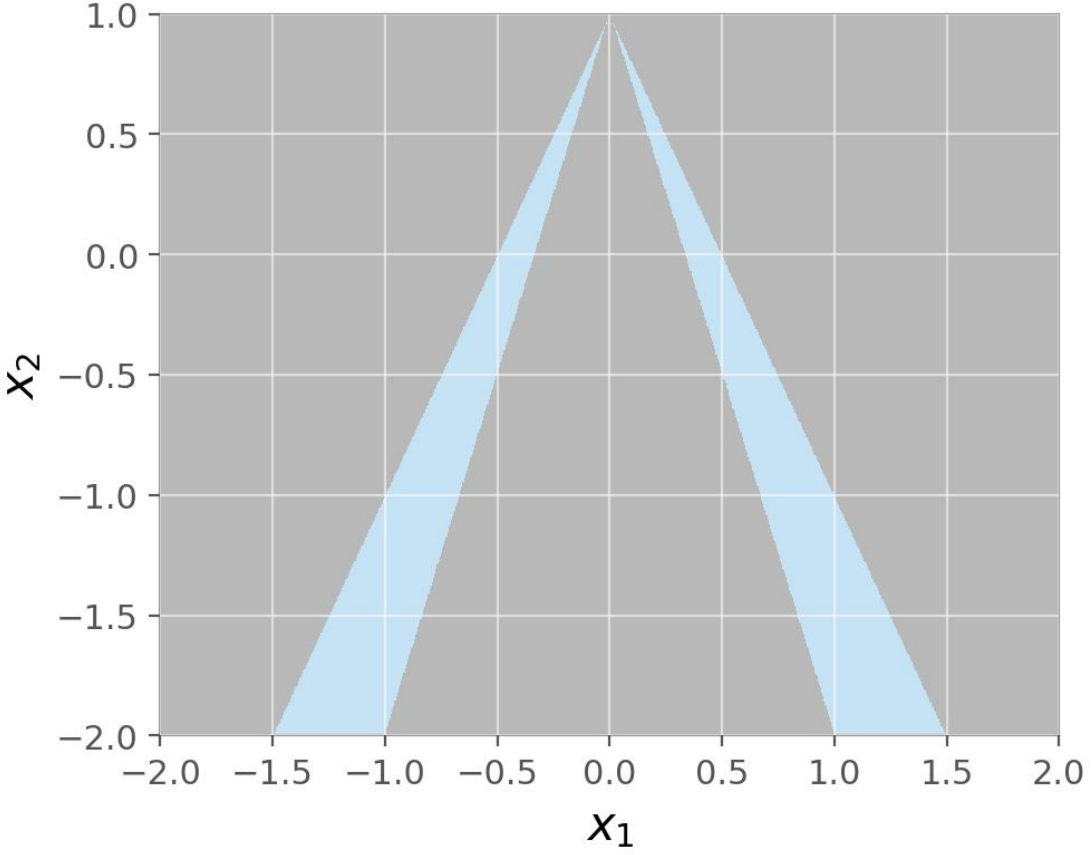


05

LIME in depth

- Intuition · algorithm · the objective · data types.
- Ribeiro, Singh & Guestrin (2016).

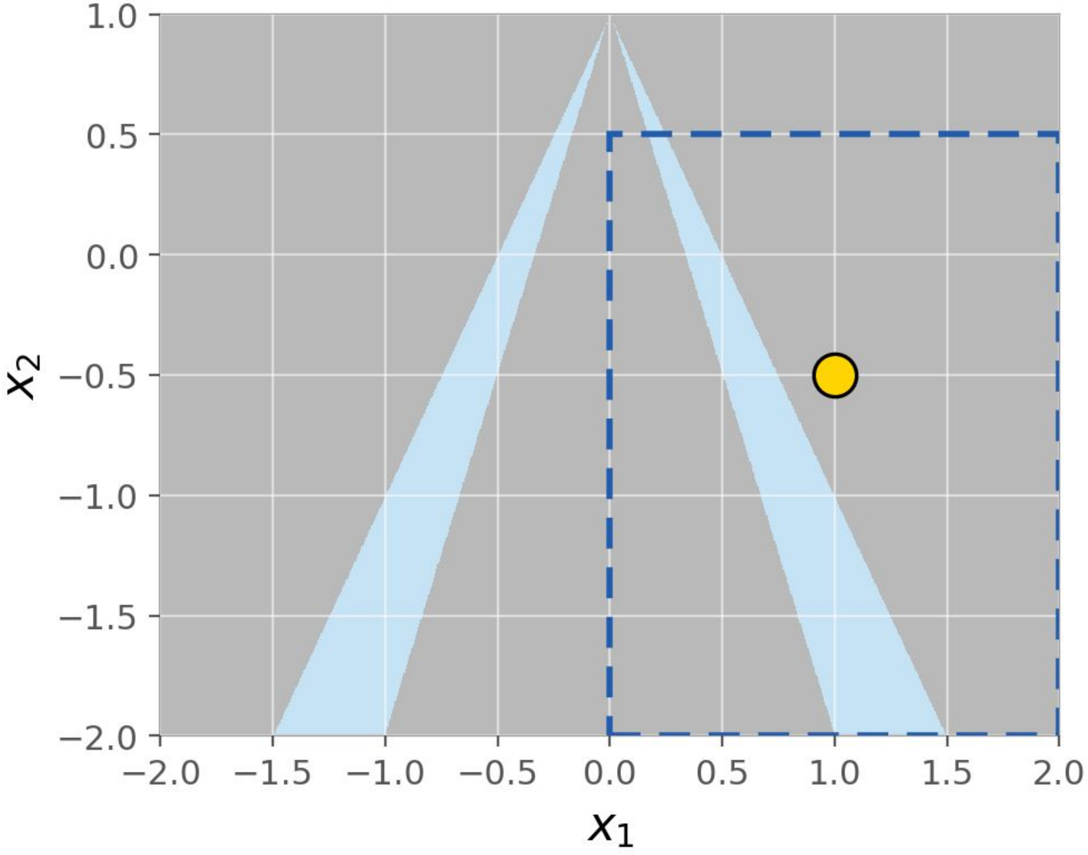
Local · Interpretable · Model-agnostic Explanations



This is **f**, the function we want to explain.

Grey is one class, blue is the other. The boundary is curved and it has two branches.

Local · Interpretable · Model-agnostic Explanations



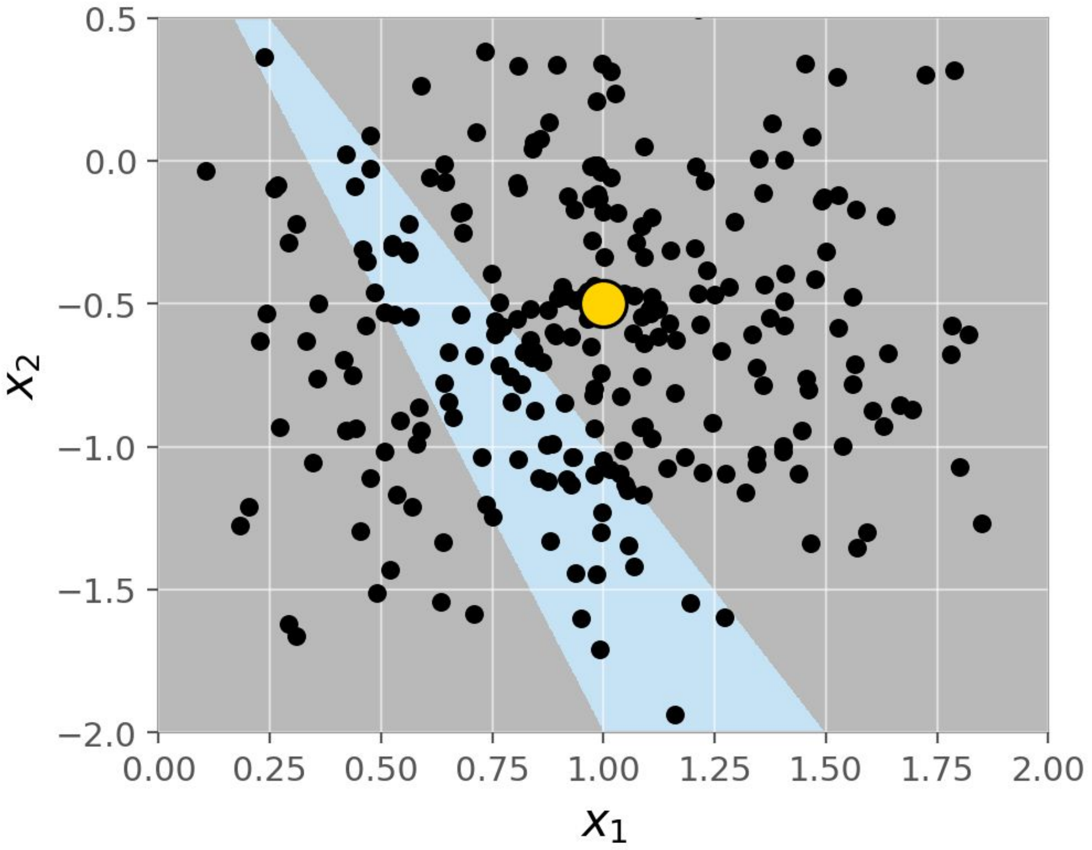
We do not explain **f**.

We explain **one prediction** the yellow point.

Everything from here happens inside the dashed box.

Local · Interpretable · Model-agnostic Explanations

Sample around it and ask f for prediction

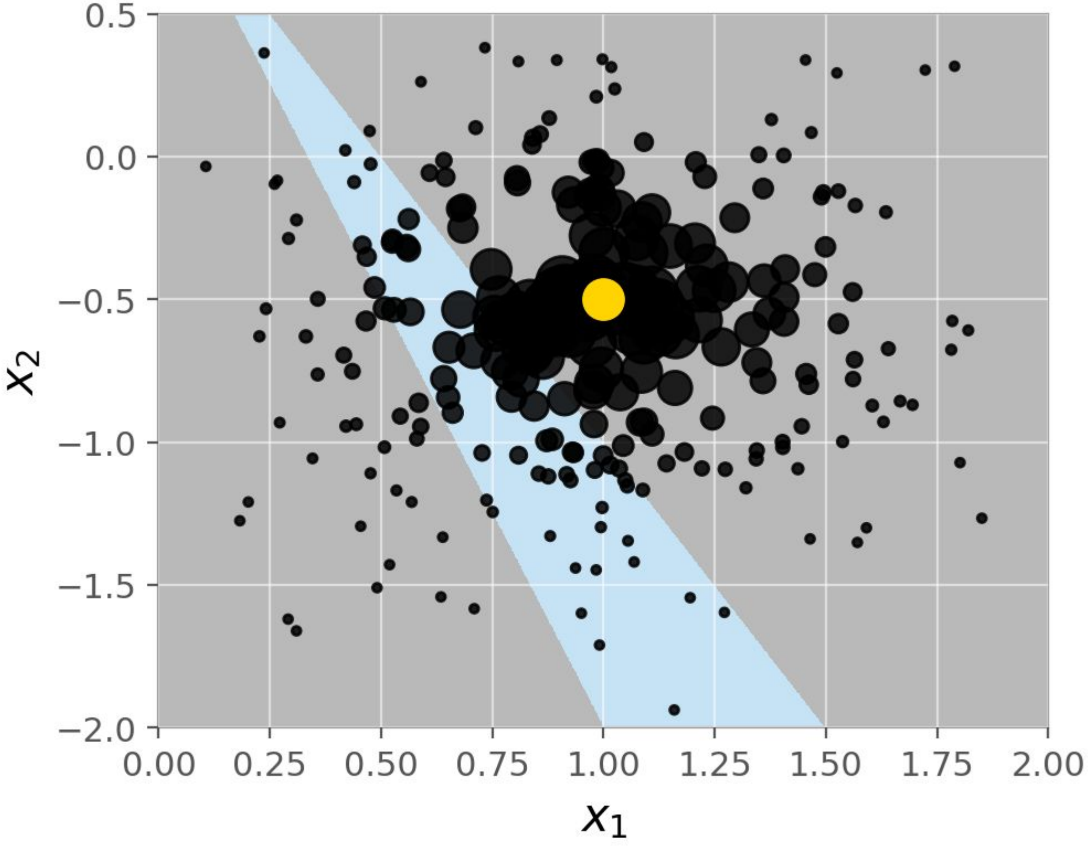


Generate many points near the instance.

Ask **f** for a label for each one.

Local · Interpretable · Model-agnostic Explanations

Weight by proximity



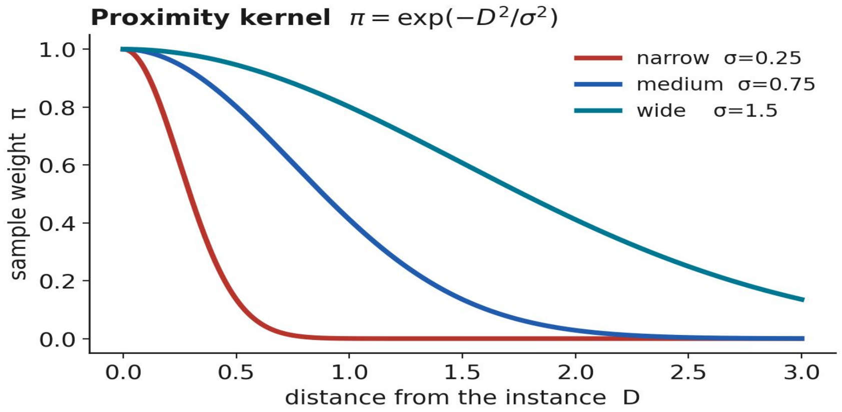
Now weight each sample by how close it is.

Big dot = counts a lot.

Small dot = almost ignored.

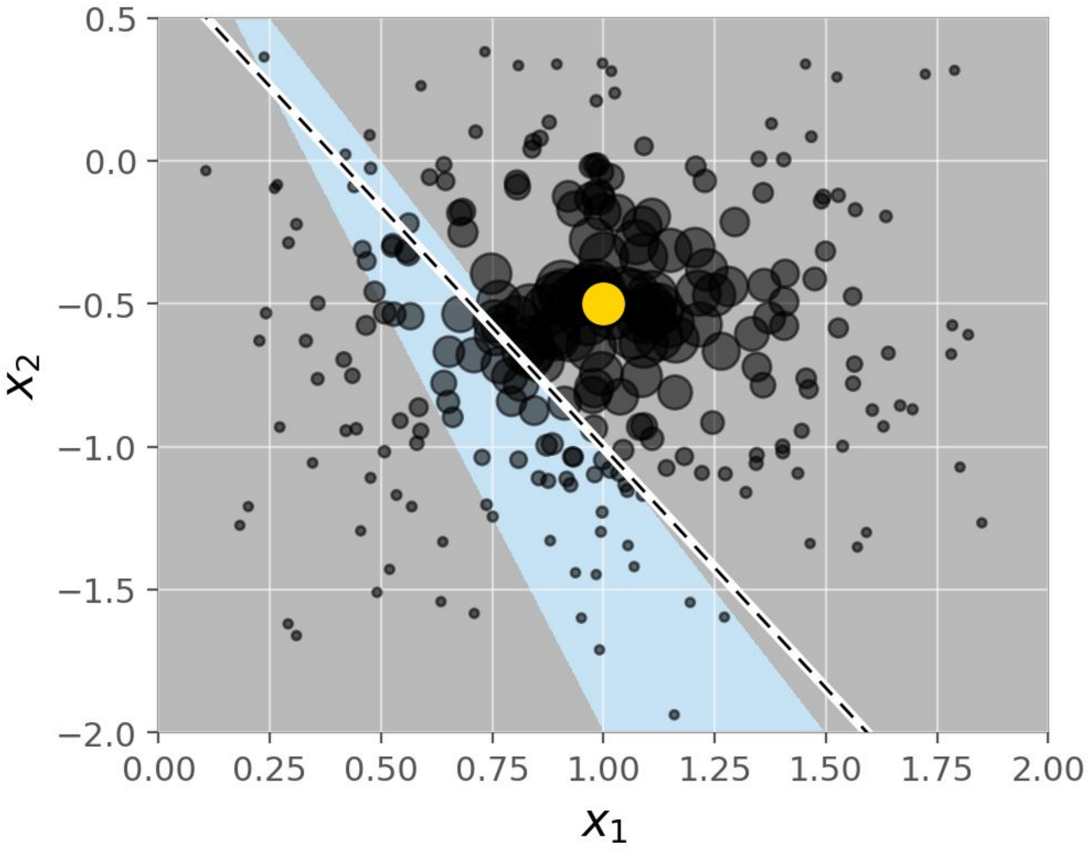
The kernel **π** decides what "near" means.

$$\pi_x(z) = \exp\left(-\frac{D(x,z)^2}{\sigma^2} \right)$$



Local · Interpretable · Model-agnostic Explanations

Fit one straight line

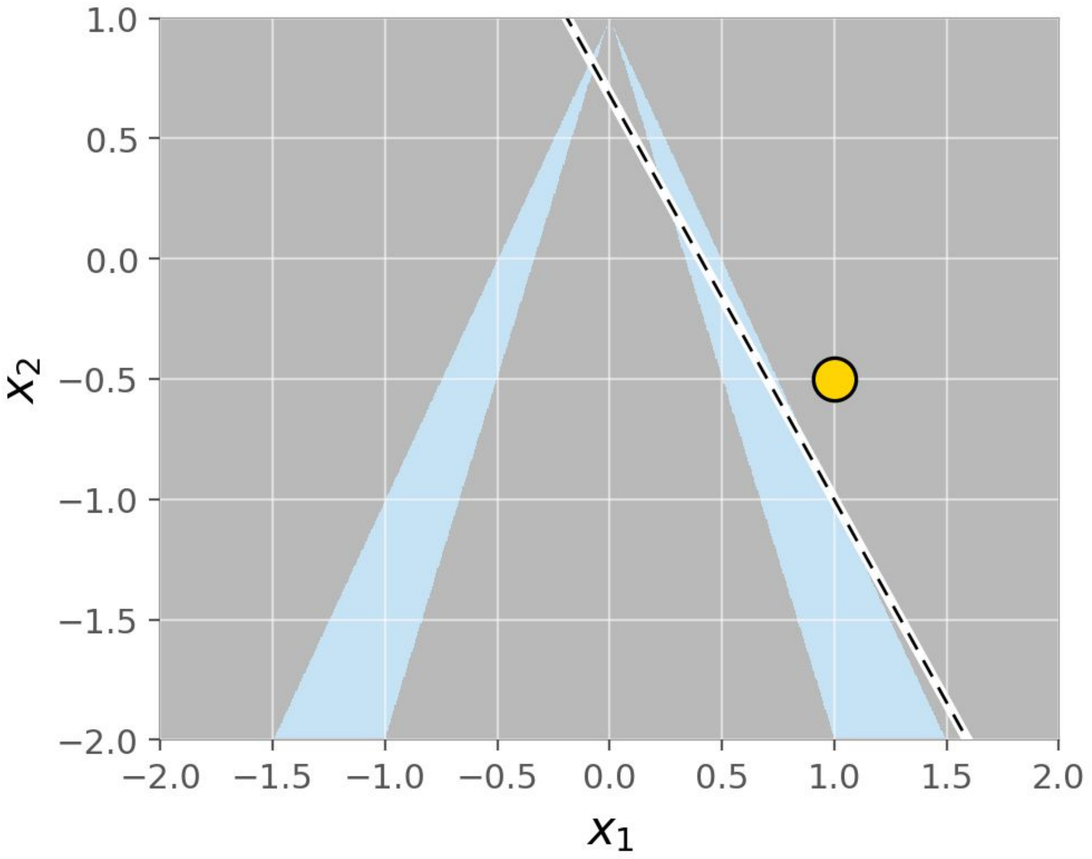


Fit a single straight line through the **weighted** samples. That line is **g**, the explanation.

Near the yellow point, the boundary looks like a straight edge. A line can copy that.

Local · Interpretable · Model-agnostic Explanations

Locally faithful. Globally wrong.



Same line. Zoomed back out.

It ignores the entire left branch. It claims half the space is grey.

As a global description of **f**, it is not accurate.

Careful: here x has 2 features, we can draw everything. A photo has no axes to draw. That is the next section.

What we are trying to do?

"By explaining a prediction, we mean presenting textual or visual artifacts that provide qualitative understanding of the relationship between the instance's components and the model's prediction."

Ribeiro, Singh & Guestrin (2016), §2

Individual

We explain one prediction of a black box model.

Not the model. One row, one photo, one sentence.

Interpretable

The explanation must be readable by a person

Faithful

It must match what the model actually does near this instance.

"interpretable" is a claim about the **reader**, not about the model.
what exactly is a person supposed to read?

Our instances

x: the raw representation, chosen for the model

Instance	Example	Raw representation — input in model f	Dimension
Text	"interpretability is the most interesting ml topic"		
Image	a photo of a wolf		
Tabular	loan applicant		

Our instances

x: the raw representation, chosen for the model

Instance	**Example**	**Raw representation — input in model f**	**Dimension**
Text	"interpretability is the most interesting ml topic"	one sentence embedding [0.0099, −1.284, … , 8.0000]	$\mathbb{R}^{768}$
Image	a photo of a wolf	pixel tensor, 3 colour channels per pixel [[[0.21, 0.19, 0.44], …]]	$\mathbb{R}^{27 \times 27 \times 3} = \mathbb{R}^{2187}$
Tabular	loan applicant	the feature values themselves [46.0, 0.66, 3.0]	$\mathbb{R}^{3}$

Interpretable data representations: from x to x′

Instance	Raw x	d	One switch =	x′	d′
Text	sentence embedding [0.0099, −1.284, … , 8.0000]	$\mathbb{R}^{768}$	one word **in this sentence** "interpretability is the most interesting ml topic"	[1,1,1,1,1,1,1]	**$\{0,1\}^7$**
Image	pixel tensor [[[0.21, 0.19, 0.44], …]]	$\mathbb{R}^{2187}$	one super-pixel	[1,1,1, … ,1]	**$\{0,1\}^{50}$**
Tabular	the three values [46.0, 0.66, 3.0]	$\mathbb{R}^3$	this feature is at its value	[1,1,1]	**$\{0,1\}^3$**

Why must x′ be all ones? Because we defined **0 = removed**. So the instance with nothing removed can only be a row of 1s.

LIME for text

Interpretable representation = word present / absent

"The plot was **predictable** and **dull**, but the **soundtrack** was genuinely **beautiful**."

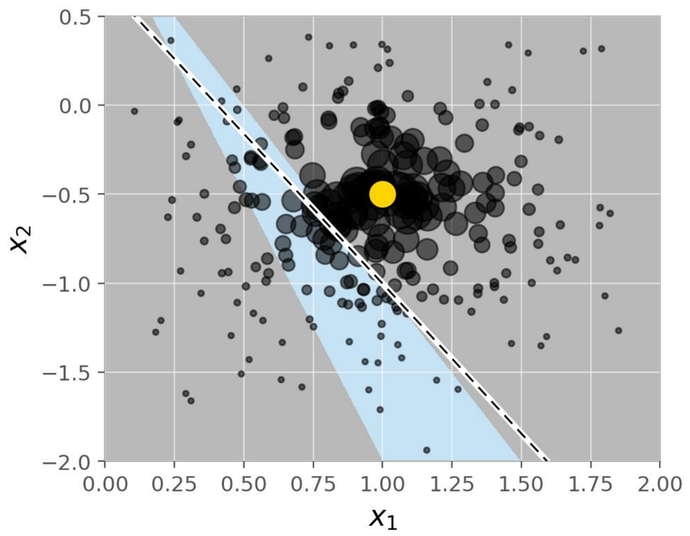
■ pushes toward NEGATIVE ■ pushes toward POSITIVE

Mechanism: toggle words on / off, re-query the model

	dull	predictable	beautiful	soundtrack
x' (original)	1	1	1	1
z'_1	0	1	1	1
z'_2	1	1	0	1
z'_3	0	1	0	1

The objective

Fidelity and interpretability, traded off



$$\xi(x) = \operatorname*{argmin}_{g \in G} \; \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

The loss

$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z')\right)^2$$

L(f , g , π_x)

Local infidelity

How far g drifts from f measured **only in the neighbourhood** that π_x defines.

Ω(g)

Complexity

How hard g is to read. For a linear g: the number of non-zero weights. **A hard cap at K**

g ∈ G

Interpretable family

Sparse linear models. We take **g(z′) = w · z′**.

Algorithm 1 sparse linear explanations

Step 1: what you must bring

```
Require:  Classifier f,  Number of samples N
Require:  Instance x,  and its interpretable version x′
Require:  Similarity kernel π□,  Length of explanation K

    Z ← { }
    for i ∈ {1, 2, 3, … , N} do
         z′□  ←  sample_around( x′ )
         Z  ←  Z  ∪  ⟨ z′□ ,  f(z□) ,  π□(z□)⟩
    end for

    w  ←  K-Lasso( Z , K )
                ▸ features: z′□    target: f(z)
    return w
```

Require: instance x, and its interpretable version x′

Algorithm 1 sparse linear explanations

Step 2 the loop builds a dataset

Applicant x: 46 kCHF · 0.66 · 3 yrs → **f(x) = 0.004** → **DENIED**

	z′ (on / off)			z (actual input to f)					
	inc	debt	yrs	inc	debt	yrs	f(z)	D	π
x'	1	1	1	46.0	0.66	3.0	**0.004**	0.00	1.00
z'_1	0	1	1	**85.1**	0.66	3.0	0.548	1.00	0.55
z'_2	1	0	1	46.0	**0.14**	3.0	**0.950**	1.00	0.55
z'_3	1	1	0	46.0	0.66	**8.8**	0.038	1.00	0.55
z'_4	0	0	1	**53.6**	**0.78**	3.0	0.004	1.41	0.31

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    end for

    w  ←  K-Lasso( Z , K )
                ▸ features: z′□    target: f(z)
    return w
```

1 = keep the applicant's value · 0 = resample it.
$\pi = \exp(-D^2/\sigma^2)$, $\sigma = 0.75\cdot\sqrt{3}$.

Algorithm 1 sparse linear explanations

Step 3 K-LASSO, then the weights

Applicant x: 46 kCHF · 0.66 · 3 yrs → **f(x) = 0.004** → **DENIED**

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    end for

    w  ←  K-Lasso( Z , K )
              ▸ features: z′□    target: f(z)
    return w
```

Row z'_2 already explains the denial on its own: resample debt to 0.14 and the model jumps **0.004** → **0.950**.

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What is returned?

w: one number per switch

w is the explanation of this one prediction

Text

w_j = the weight of word j.
Sign = direction, size = strength.
Show the K largest $|w_j|$.

Image

w_j = the weight of super-pixel j.
Highlight the K with the largest positive weight, grey out the rest.

Tabular

w_j = the effect of **keeping** feature j at his value.
Label the bar with the value.

What is LIME?

Local Interpretable Model-agnostic Explanations

Local surrogate models are interpretable models used to explain individual predictions of black-box models. LIME (Ribeiro, Singh & Guestrin, 2016) fits such a surrogate trained to approximate the black box's predictions in the neighborhood of one instance.

L **Local**

Explains one prediction, in its neighborhood and not the whole model.

I **Interpretable**

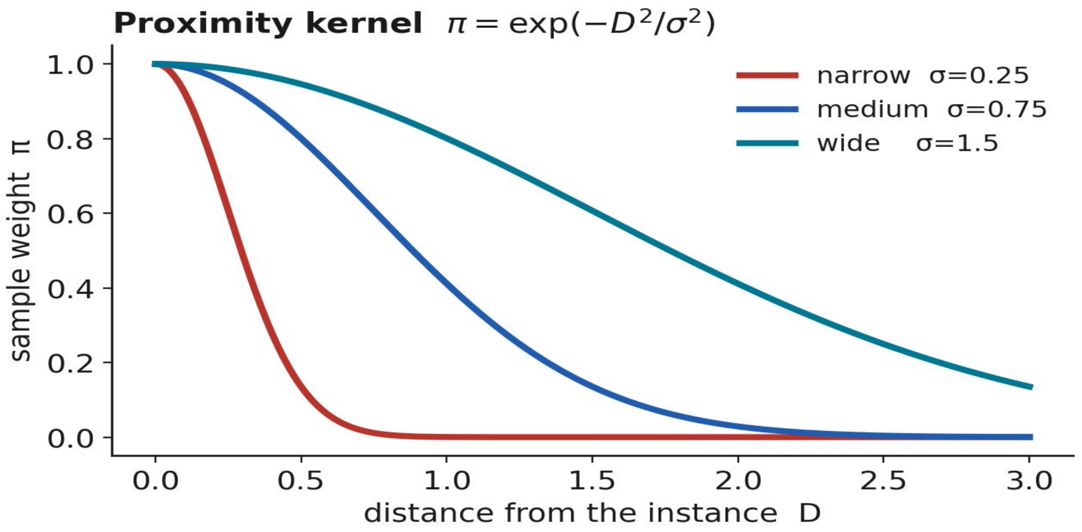
The surrogate is a simple, sparse model a human can read.

M **Model-agnostic**

Treats the black box as a query-only oracle.

Proximity Kernel

Same objective, three kernel widths → different notions of “local”.



π_x defines “nearby”. The proximity kernel decides which samples count

LIME for text: which words drove it?

Interpretable representation = word present / absent

"The plot was **predictable** and **dull**, but the **soundtrack** was genuinely **beautiful**."

■ pushes toward NEGATIVE ■ pushes toward POSITIVE

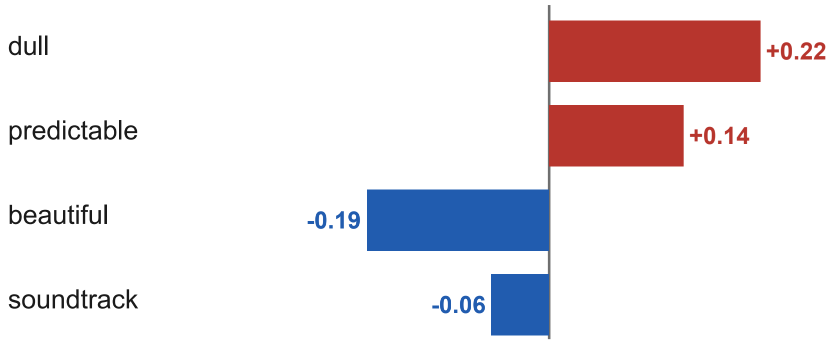
Prediction
NEGATIVE
p = 0.63

Mechanism: toggle words on / off, re-query the model

	dull	predictable	beautiful	soundtrack	p(neg)
x' (original)	1	1	1	1	**0.63**
z'_1 − dull	0	1	1	1	**0.41**
z'_2 − beautiful	1	1	0	1	**0.79**
z'_3 − both	0	1	0	1	**0.55**

Remove "dull" → less negative. Remove "beautiful" → more negative.

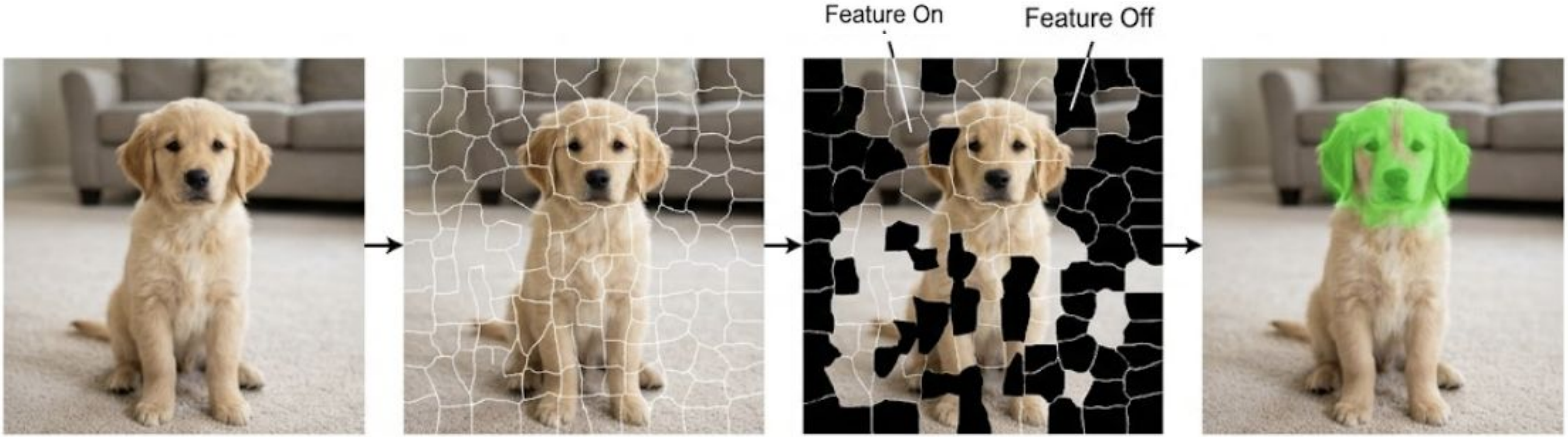
The explanation: word weights



+ toward NEGATIVE · − toward POSITIVE

LIME for images: turn super-pixels on / off

The interpretable representation is a binary super-pixel mask



1 Original

the raw image the black box scored.

2 Super-pixels

group neighbouring pixels into ~40 patches, the on/off units.

3 Perturb

grey out a random subset (1 = shown, 0 = greyed) re-query the model.

4 Explain

patches whose presence most raised the score = the explanation.

LIME in numbers: one explanation, end to end

A real gradient-boosting loan model · one applicant · every step with actual numbers

1 **Applicant x:** income **46 kCHF** · debt ratio **0.66** · **3** yrs employed → *black box f* → **f(x) = 0.004** → DENIED

2 Perturb → query → weight

5 000 samples drawn 5 shown

	z′ (on / off)			z (actual input to f)					
	inc	debt	yrs	inc	debt	yrs	f(z)	D	π
x′	1	1	1	46.0	0.66	3.0	0.004	0.00	1.00
z'_1	0	1	1	85.1	0.66	3.0	0.548	1.00	0.55
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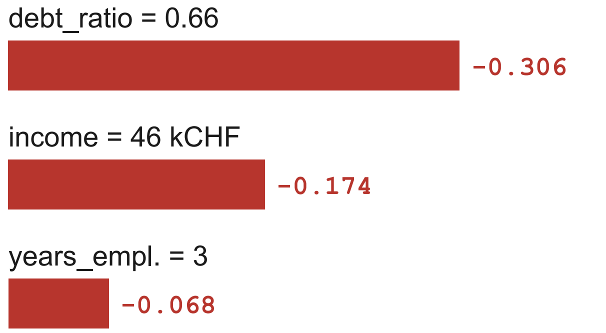
1 = keep the applicant's value · 0 = resample it · $\pi = \exp(-D^2/\sigma^2)$, $\sigma = 0.75\cdot\sqrt{3}$

3 Fit → the explanation

weighted ridge on z′ → coefficients



All three negative — each one is holding this application down.

Each coefficient = the effect of ***keeping*** that feature at the applicant's value. Debt ratio 0.66 is the single biggest reason for the denial (−0.31)
row z'_2 shows it directly: resample debt to 0.14 and the model jumps 0.004 → 0.950.

06

Limitations of LIME

- How local is “local”? · instability.
- The cracks that motivate SHAP and robustness.

Kernel width: how local is "local"?

The proximity kernel $\pi = \exp(-D^2/\sigma^2)$

σ too small → the line chases the local tangent

σ too large distant points pull the line → it spans a curved region → local linearity breaks, signs can flip.

No principled choice tabular default ≈ $0.75 \cdot \sqrt{(\#features)}$
A heuristic, not a derivation.

The robustness problem: the same point yields qualitatively different explanations and there's no ground truth to arbitrate.

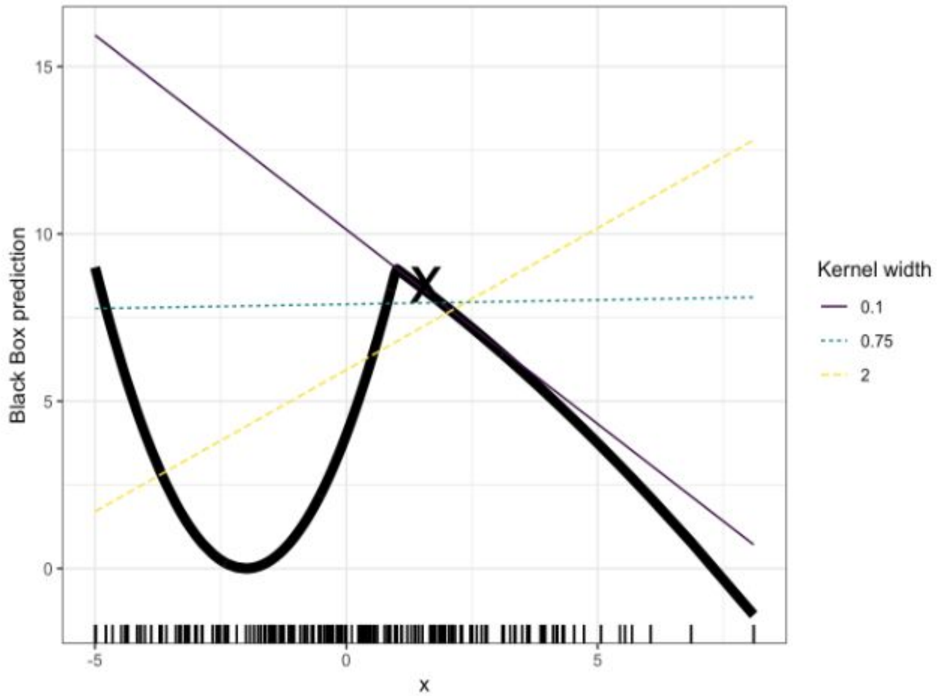


Figure 14.2: Explanation of the prediction of instance x = 1.6 with different kernel widths. The model predictions are shown as a thick line and the distribution of the data is shown with rugs. Three local surrogate models with different kernel widths are computed.

Far-away points, instability & other limits

The far-away tension

- By design the kernel makes far points near-weightless
- But widen the neighborhood and those same far points misrepresent the local gradient.
- “Local” and “faithful” pull against each other.

Other practical limits

Instability: random sampling → two runs give different explanations for the same point.

Neighborhood: defining it for tabular data is ad-hoc.

Correlations: perturbations can create unrealistic feature combinations.

Manipulable: explanations can be adversarially fooled

Bridge to SHAP

LIME’s open questions

- How wide should the neighborhood be? (kernel width)
- Why these weights, and are they consistent?
- Will I get the same answer twice?

SHAP’s answer

- Attributions grounded in game theory (Shapley values).
- Additive, with consistency & local-accuracy guarantees.
- Replaces several of LIME’s ad-hoc choices with axioms.

07

From local to global: SP-LIME

- Two trust problems.
- A global picture assembled from local pieces.

Two trust problems

? Trust a prediction

- “Should I act on this individual output?”
- Solved by one local, faithful, sparse explanation.

→ this is LIME.

? Trust the model

- “Will it behave reasonably once deployed?”
- You can’t read thousands of local explanations.
- Users have a limited budget B of explanations to inspect.

→ this needs SP-LIME.

SP-LIME: submodular pick

A representative handful of local explanations

Goal: pick B instances whose explanations are maximally covering and non-redundant.

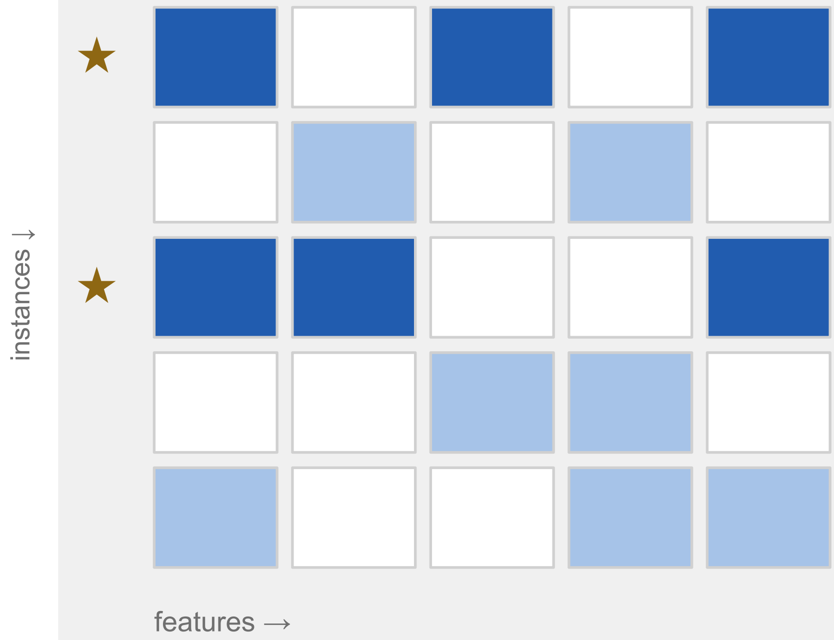
How: frame it as a submodular coverage problem: reward covering important features not yet seen.

Solve: greedy selection, with the classic 1 − 1/e approximation guarantee.

Crucial: each explanation stays strictly ***local***.

“Global” means a curated ***collection*** of local explanations

Pick rows that cover the most (weighted) columns



★ picked instances cover the important features with no overlap.

08

LIME today

- Can we trust the explanation itself?
- From attacks to provable guarantees.

How LIME is used today

From research idea to everyday XAI tool

M

Model debugging

Surface spurious shortcuts and data leakage before shipping.

N

NLP

Token-level attributions for text classifiers and, increasingly, LLM outputs.

V

Vision

Super-pixel explanations for image models what region drove the label.

R

Regulated domains

Per-decision justification in credit, insurance, healthcare.

LIME in Applied Domains

From research idea to everyday XAI tool

Classification and Detection
Model Prediction Analysis

What you are seeing: LIME highlights the parts of the image (superpixels) most strongly contributing to the model's detection in green and the rest in red. LIME here helps us validate that the model is indeed looking at the features of the dog when predicting 'dog' and not at the cat in the background.

https://github.com/marcotcr/lime

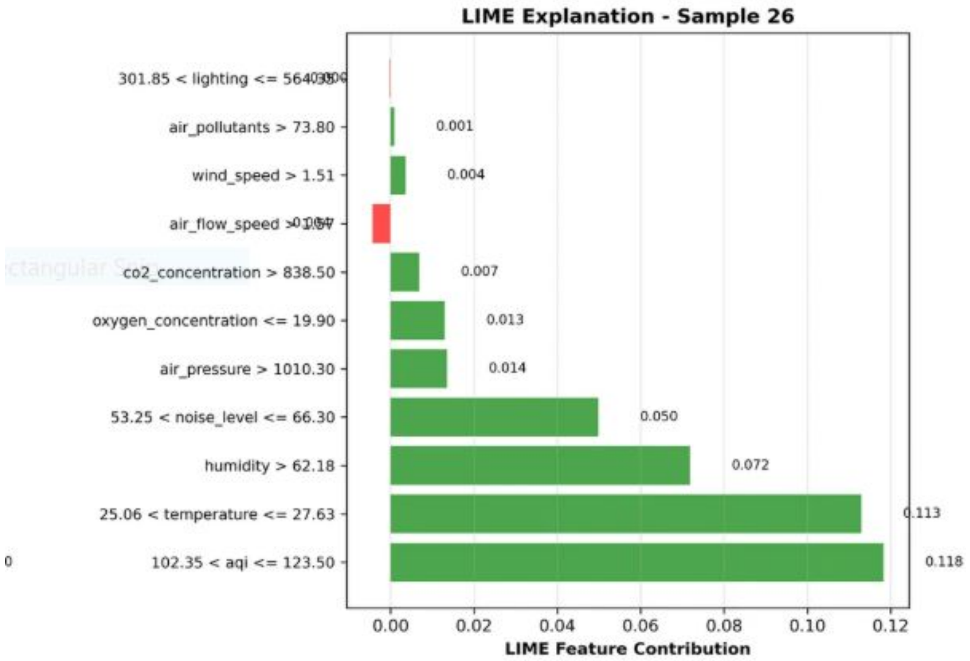
Detection model predicted 'Dog' for this image.

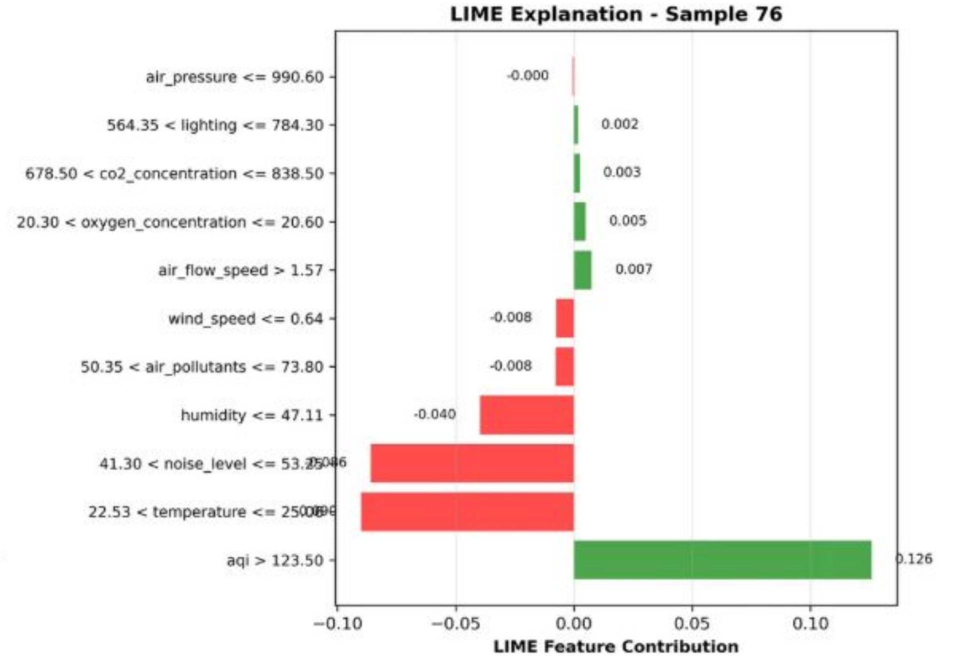
LIME in Applied Domains

From research idea to everyday XAI tool

Machine learning prediction model for medical environment comfort

What you are seeing: An XGBoost model has been trained for medical environment comfort prediction. LIME explanation shows contributions of various features to prediction results under different environmental conditions. Green bars represent features that increase discomfort probability, and red bars represent features that decrease discomfort probability. In Sample 26 all main features are showing positive contributions, indicating that this sample faces multiple environmental risk factors. However, for Sample 76, AQI is the only determining factor for discomfort, while other factors like humidity, noise, and temperature are contributing towards a comforting environment.



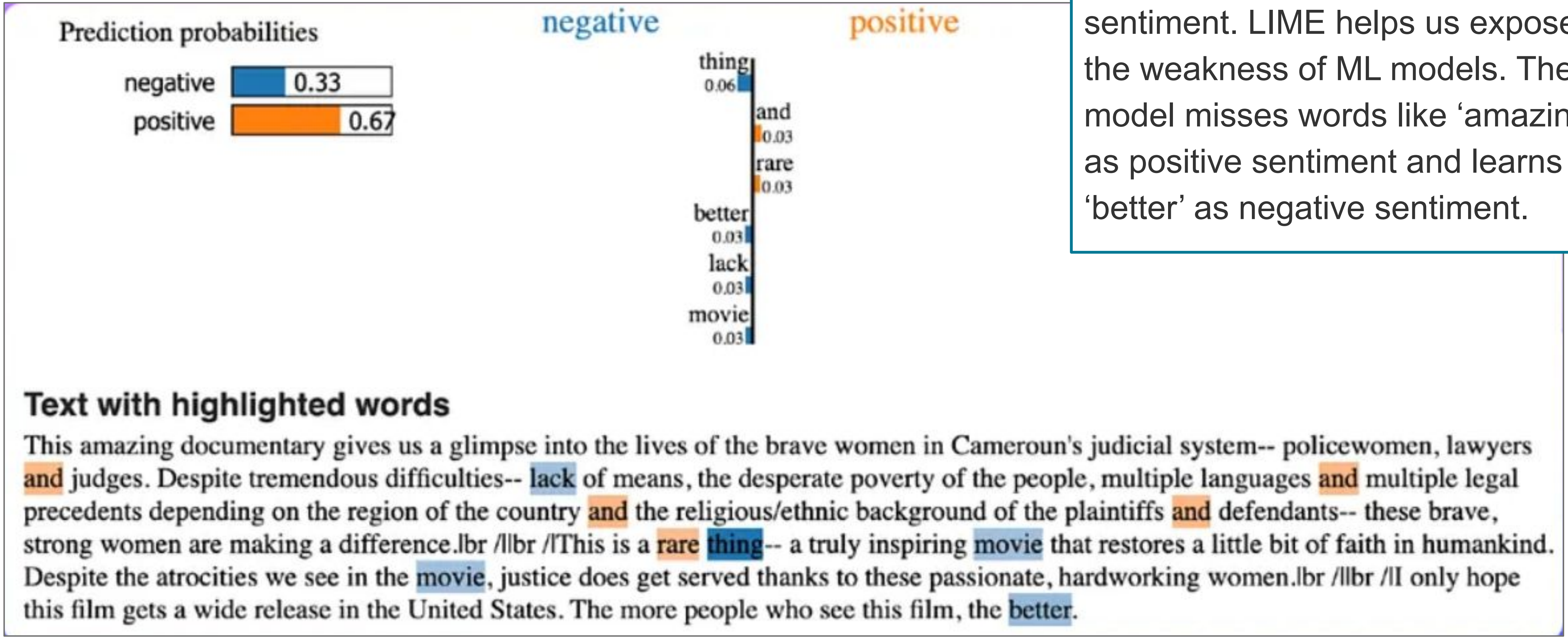


Zhang, C., Liu, L. Machine learning prediction model for medical environment comfort based on SHAP and LIME interpretability analysis. Sci Rep ***15****, 39269 (2025).*

LIME in Applied Domains

From research idea to everyday XAI tool

Sentiment Analysis of Movie Reviews



What you are seeing: LIME explanation to highlight words which showcase positive vs negative sentiment. LIME helps us expose the weakness of ML models. The model misses words like 'amazing' as positive sentiment and learns 'better' as negative sentiment.

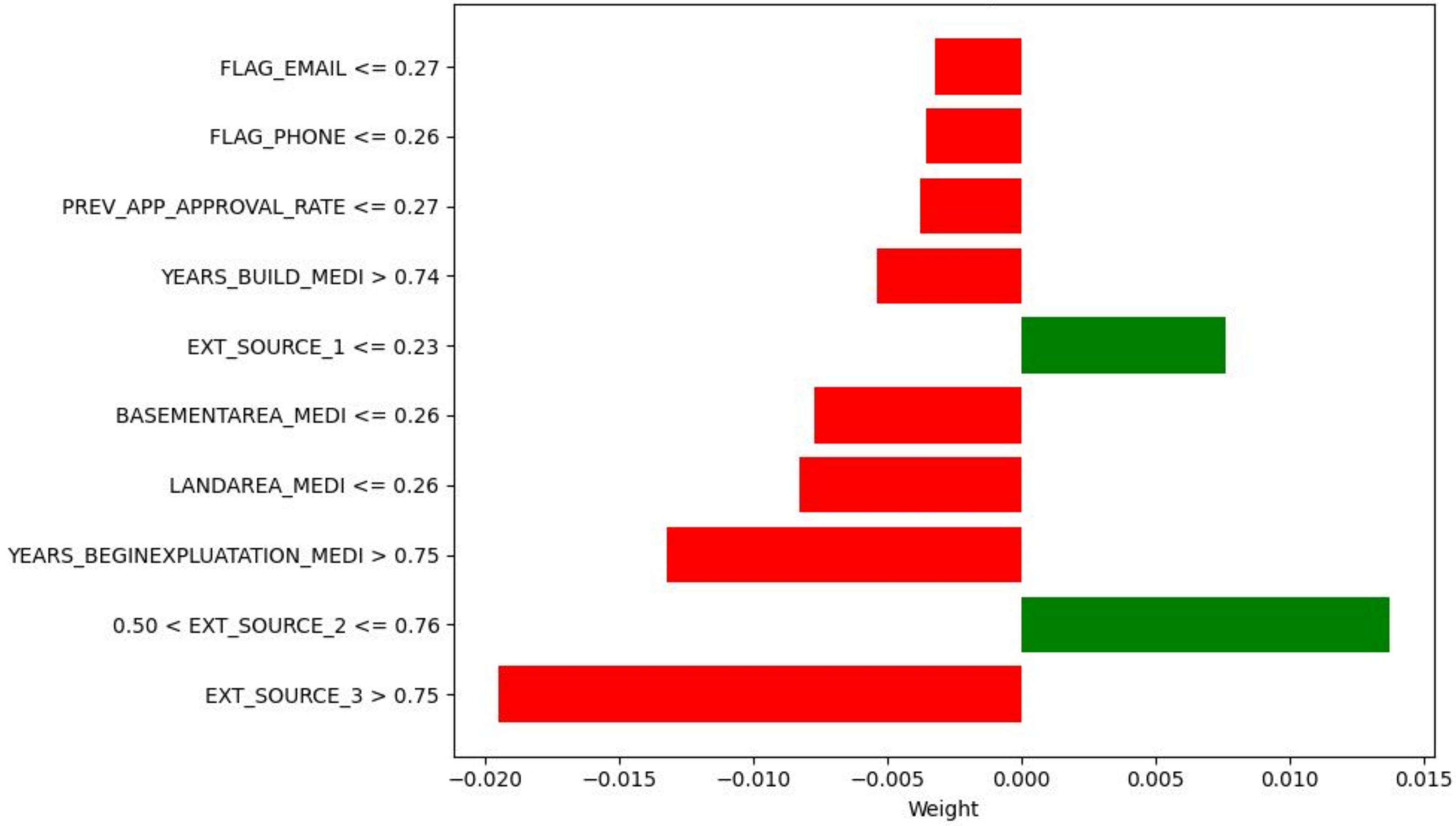
Chiusano, Fabio. "Two minutes NLP— Explain predictions with LIME." Generative AI, 21 Feb. 2022, medium.com/nlplanet/two-minutes-nlp-explain-predictions-with-lime-aec46c7c25a2.

LIME in Applied Domains

From research idea to everyday XAI tool

Explainable Artificial Intelligence Credit Risk Assessment using ML

LIME feature importance of high-risk applicant



What you are seeing: LIME Feature Importance gives insights into specific applications providing users transparency about the model's decision for each applicant during their risk report generation. This provides decision-makers, as well as applicants, ability to review their position and reasoning.

Shreya, and Harsh Pathak. "Explainable Artificial Intelligence Credit Risk Assessment using Machine Learning." arXiv, 2025, https://arxiv.org/abs/2506.19383.

Takeaways

1	**Right answer ≠ right reason**	High accuracy hides Clever Hans shortcuts. We need the “why”, not just the “what”.
2	**LIME = local surrogate**	Perturb the interpretable representation locally, weight by proximity, fit a sparse model. Faithful locally, silent globally.
3	**Local → global via SP-LIME**	A representative, non-redundant handful of local explanations lets even non-experts pick the better model.
4	**Fragility → robustness & certification**	Kernel width, instability and adversarial attacks

Thank you

Questions & discussion

Key references

Ribeiro, Singh & Guestrin (2016) “Why Should I Trust You?”, KDD. · Molnar, Interpretable Machine Learning.
Slack et al. (2020), Fooling LIME and SHAP, AIES. · Alvarez-Melis & Jaakkola (2018). · Jin et al. (2025); Xue et al. (2023), MuS.